

A DEGREE-18 COUNTEREXAMPLE TO CONJECTURE 5.23 OF ACEVEDO–BLEKHERMAN–DEBUS–RIENER

ABSTRACT. Acevedo, Blekherman, Debus, and Riener conjectured that $\text{trop}(\mathcal{BS}_{2d}^*) = T_{2d}^{(2)}$ for every $d \geq 3$. We disprove their Conjecture 5.23 by giving an explicit vector

$$y \in \text{trop}(\mathcal{BS}_{18}^*) \setminus T_{18}^{(2)}.$$

In exact integer arithmetic, y satisfies all 417 first-order superdominance inequalities on the even partitions of 18 and all 1056 inequalities generated by the degree-9 partial-symmetry description; by the membership criterion recalled below, this places y in the tropicalized sums-of-squares cone. One second-order superdominance inequality has slack -2 .

1. THE CONJECTURE AND A FINITE TEST

We use the notation of Acevedo, Blekherman, Debus, and Riener [1]. Let Λ_{2d}^{ev} be the set of even partitions of $2d$. Write the parts of a partition increasingly, $\lambda_{(1)} \leq \dots \leq \lambda_{(\ell(\lambda))}$. For partitions of the same integer, define superdominance by

$$\lambda \succeq \mu \iff \sum_{i=1}^j \lambda_{(i)} \leq \sum_{i=1}^j \mu_{(i)} \quad \text{for } 1 \leq j \leq \min\{\ell(\lambda), \ell(\mu)\}.$$

If $\lambda \mu$ denotes multiset union of parts, followed by sorting, and $\mu^{\circ 2} = \mu \mu$, then

$$(1) \quad T_{2d}^{(2)} = \left\{ y \in \mathbb{R}^{\Lambda_{2d}^{\text{ev}}} : y_{\lambda^1} + y_{\lambda^2} \geq 2y_{\mu} \text{ whenever } \lambda^1 \lambda^2 \succeq \mu^{\circ 2} \right\}.$$

Define also

$$T_{2d}^{(1)} = \left\{ y \in \mathbb{R}^{\Lambda_{2d}^{\text{ev}}} : y_{\lambda} \geq y_{\mu} \text{ whenever } \lambda \succeq \mu \right\}.$$

The source proves $T_{2d}^{(1)} \supseteq \text{trop}(\mathcal{BS}_{2d}^*) \supseteq T_{2d}^{(2)}$ and conjectures the second inclusion to be equality for all $d \geq 3$ [1, Propositions 5.13 and 5.20 and Conjecture 5.23]; all numbering here is that of arXiv:2408.04616v1.

We record the finite partial-symmetry test used below. For $r, s \geq 0$, let $\mathcal{V}_{r,s}(d)$ consist of pairs $u = (\alpha, \lambda)$, where

$$\alpha = (\alpha_1, \dots, \alpha_{r+s}) \in \mathbb{Z}_{>0}^{r+s},$$

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the first r entries of α are odd, the last s are even, λ is an even partition, and $|\alpha| + |\lambda| = d$. Since $|\alpha| \equiv r \pmod{2}$ and $|\lambda|$ is even, the parity constraint forces $r \equiv d \pmod{2}$, and $r + 2s \leq d$; for $d = 9$ this leaves r odd and 15 nonempty blocks. All lists of parts below are sorted into partitions. For $u = (\alpha, \lambda)$, set

$$A_u = (2\alpha)\lambda\lambda, \quad B_u = (2\alpha)^*\lambda\lambda \quad (r + s \geq 2),$$

where $(2\alpha)^*$ is obtained from (2α) by merging its two largest parts. The restriction $r + s \geq 2$ is not a choice: for $r + s = 1$ the monomial symmetric function $m_{2\alpha} = m_{(2\alpha_1)}$ is the power sum $p_{2\alpha_1}$, whose negative support is empty, so single-part blocks contribute no type-(1) inequality. These are the $131 - 119 = 12$ terms with $r + s = 1$ counted in the proof of Theorem 2. For $u = (\alpha, \lambda)$ and $v = (\beta, \mu)$ in the same block, set

$$C_{u,v} = (\alpha + \beta)\lambda\mu,$$

where $\alpha + \beta$ is componentwise.

Lemma 1 (Degree-18 membership test). *Let $H \subseteq \mathbb{R}^{\Lambda_{18}^{\text{sy}}}$ be the set of vectors z satisfying*

$$(2) \quad z_{A_u} \geq z_{B_u} \quad (u \in \mathcal{V}_{r,s}(9), \ r + s \geq 2), \quad z_{A_u} + z_{A_v} \geq 2z_{C_{u,v}} \quad (u \neq v)$$

over all nonempty blocks $\mathcal{V}_{r,s}(9)$. Then

$$\text{trop}(\mathcal{B}\Sigma_{18}^*) = T_{18}^{(1)} \cap H.$$

Repeated inequalities are harmless.

Proof. We prove $T_{18}^{(1)} \cap H \subseteq \text{trop}(\mathcal{B}\Sigma_{18}^*)$ first, so let $z \in T_{18}^{(1)} \cap H$. For each (r, s) , form the row vector of all terms

$$n^{(r+s)/2} X_1^{\alpha_1} \cdots X_{r+s}^{\alpha_{r+s}} p_\lambda, \quad (\alpha, \lambda) \in \mathcal{V}_{r,s}(9).$$

These rows span the degree-9 forms after the partial-symmetry decomposition. The limiting Gram blocks in [1, Section 5.3 and Appendix A] have, up to positive scalar factors, entries

$$m_{2\alpha} p_\lambda^2 \quad \text{and} \quad m_{\alpha+\beta} p_\lambda p_\mu$$

on the diagonal and off the diagonal, respectively, where m_γ is a monomial symmetric function. Lemma 5.17 and Corollary 5.18 of [1] identify the superdominance-maximal indices in the positive support, the negative support, and the full support of $m_{2\alpha}$, respectively $m_{\alpha+\beta}$. Those corollaries concern $m_{2\alpha}$ alone, whereas the indices occurring in the Gram entries are $(2\alpha)\lambda\lambda$ and $(\alpha + \beta)\lambda\mu$; the two are matched as follows. The factor p_λ^2 (resp. $p_\lambda p_\mu$) contributes the same multiset $\lambda\lambda$ (resp. $\lambda\mu$) of parts to every index of the entry, so the merges of Lemma 5.17 take place only inside $m_{2\alpha}$ (resp. $m_{\alpha+\beta}$) and the indices being compared differ only in that factor. The maximality of $(2\alpha)^*$ in the negative support of $m_{2\alpha}$ therefore transports to $(2\alpha)^*\lambda\lambda = B_u$, and in the same way the positive diagonal, negative diagonal, and full off-diagonal maxima are A_u , B_u , and $C_{u,v}$. We note a notational

divergence: the proof of [1, Proposition 5.20] writes the type-(1) inequality coming from a diagonal entry as $y_\nu \geq y_{\nu^*}$ with $\nu = \alpha\lambda$, starring the whole index, whereas B_u above stars (2α) only. Because $z \in T_{18}^{(1)}$, the coordinate maxima are attained at those same indices, so (2) verifies the max inequalities of [1, Lemma 5.14]. Those inequalities are quantified over all pairs of rows, whereas (2) imposes an off-diagonal condition only for $u \neq v$ lying in the same block; for rows in different blocks the entry of the limiting Gram matrix vanishes, so the associated index set is empty and the condition is vacuous under the convention $\max \emptyset = -\infty$ of [1, Lemma 5.14]. The full-dimensional closure argument in the proof of [1, Proposition 5.19] then places z in $\text{trop}(\mathcal{B}\Sigma_{18}^*)$.

For the reverse inclusion, $\text{trop}(\mathcal{B}\Sigma_{18}^*) \subseteq T_{18}^{(1)}$ by [1, Proposition 5.13], and every point of $\text{trop}(\mathcal{B}\Sigma_{18}^*)$ satisfies the max inequalities of [1, Lemma 5.14]. On $T_{18}^{(1)}$ the maxima in those inequalities are attained at A_u , B_u , and $C_{u,v}$, as above, so there they reduce to (2). Hence $\text{trop}(\mathcal{B}\Sigma_{18}^*) \subseteq T_{18}^{(1)} \cap H$. $\square$

2. THE COUNTEREXAMPLE

Define $y \in \mathbb{R}^{\Lambda_{18}^{\text{ev}}}$ by the following table; exponential notation denotes repeated parts.

	λ	y_λ	λ	y_λ	λ	y_λ
	(18)	0	(10, 4, 2 ²)	24	(6 ² , 2 ³)	36
	(16, 2)	9	(10, 2 ⁴)	44	(6, 4 ³)	20
	(14, 4)	6	(8 ² , 2)	12	(6, 4 ² , 2 ²)	30
	(14, 2 ²)	20	(8, 6, 4)	10	(6, 4, 2 ⁴)	48
(3)	(12, 6)	3	(8, 6, 2 ²)	24	(6, 2 ⁶)	68
	(12, 4, 2)	14	(8, 4 ² , 2)	21	(4 ⁴ , 2)	30
	(12, 2 ³)	32	(8, 4, 2 ³)	36	(4 ³ , 2 ³)	40
	(10, 8)	0	(8, 2 ⁵)	56	(4 ² , 2 ⁵)	60
	(10, 6, 2)	12	(6 ³)	10	(4, 2 ⁷)	80
	(10, 4 ²)	12	(6 ² , 4, 2)	20	(2 ⁹)	100

Theorem 2. *The vector y in (3) satisfies*

$$y \in \text{trop}(\mathcal{B}\Sigma_{18}^*) \setminus T_{18}^{(2)}.$$

Consequently, Conjecture 5.23 of [1] is false and

$$\text{trop}(\mathcal{B}\Sigma_{18}^*) \supsetneq T_{18}^{(2)}.$$

Proof. For $d = 9$, there are 15 nonempty blocks $\mathcal{V}_{r,s}(9)$, containing 131 terms in total. There are 417 nontrivial first-order superdominance inequalities on Λ_{18}^{ev} , and exact substitution gives minimum slack 0 over all of them. The vector y therefore belongs to $T_{18}^{(1)}$. The system (2) consists of 119 diagonal and 937 off-diagonal inequalities, hence 1056 generated inequalities; exact

substitution again gives minimum slack 0, so $y \in H$. By Lemma 1,

$$y \in \text{trop}(\mathcal{B}\Sigma_{18}^*).$$

Now take

$$\lambda^1 = (10, 8), \quad \lambda^2 = (4, 4, 4, 2, 2, 2), \quad \mu = (8, 4, 4, 2).$$

The increasingly ordered prefix sums of $\lambda^1\lambda^2$ and $\mu^{\circ 2}$ are

$$(2, 4, 6, 10, 14, 18, 26, 36) \quad \text{and} \quad (2, 4, 8, 12, 16, 20, 28, 36),$$

respectively. Hence $\lambda^1\lambda^2 \succeq \mu^{\circ 2}$, so (1) requires

$$y_{(10,8)} + y_{(4,4,4,2,2,2)} \geq 2y_{(8,4,4,2)}.$$

Its slack at (3) is

$$0 + 40 - 2 \cdot 21 = -2.$$

Therefore $y \notin T_{18}^{(2)}$, as claimed. $\square$

Exact verification. The program reproduced in the appendix uses only the Python standard library and no floating-point arithmetic. It generates all partitions, blocks, and inequalities from the definitions above, checks the counts 15, 131, 119, 937, 1056, verifies all 417 first-order superdominance inequalities and all partial-symmetry inequalities with minimum slack 0, and verifies the second-order violation with slack -2 . It records two further facts about the finite test of Lemma 1: each of the 119 diagonal inequalities has $A_u \succeq B_u$ and is therefore already a first-order inequality, and each of the 937 off-diagonal ones has $A_u A_v \succeq C_{u,v}^{\circ 2}$ and is therefore itself an instance of (1). Finally, of the 7025 inequalities of (1) on Λ_{18}^{ev} —one for each unordered pair $\{\lambda^1, \lambda^2\}$, repetition allowed, together with each μ satisfying $\lambda^1\lambda^2 \succeq \mu^{\circ 2}$ —exactly one is violated by y , namely $y_{(10,8)} + y_{(4,4,4,2,2,2)} \geq 2y_{(8,4,4,2)}$.

APPENDIX A. THE VERIFICATION PROGRAM

The following self-contained program performs the checks described above. Comments give the expected output, one `print` statement per line.

```
from itertools import combinations
from itertools import combinations_with_replacement as cwr

def parts(n, mx=None):
    # partitions, parts decreasing
    mx = n if mx is None else mx
    if n == 0:
        yield ()
        return
    for k in range(min(n, mx), 0, -1):
        for t in parts(n - k, k):
            yield (k,) + t

def evens(m):
    # even partitions of m
    return [] if m % 2 else [tuple(2*p for p in t)
                             for t in parts(m//2)]
```

```

def key(ls):
    return tuple(sorted(ls, reverse=True))

def sd(a, b):
    # a superdominates b
    A, B, sa, sb = sorted(a), sorted(b), 0, 0
    for i in range(min(len(A), len(B))):
        sa, sb = sa + A[i], sb + B[i]
        if sa > sb:
            return False
    return True

def alphas(r, s, m):
    # ordered: r odd, then s even
    if r:
        for a in range(1, m + 1, 2):
            for t in alphas(r - 1, s, m - a):
                yield (a,) + t
    elif s:
        for a in range(2, m + 1, 2):
            for t in alphas(0, s - 1, m - a):
                yield (a,) + t
    else:
        yield ()

Y = {(18,):0,(16,2):9,(14,4):6,(14,2,2):20,(12,6):3,
      (12,4,2):14,(12,2,2,2):32,(10,8):0,(10,6,2):12,(10,4,4):12,
      (10,4,2,2):24,(10,2,2,2,2):44,(8,8,2):12,(8,6,4):10,
      (8,6,2,2):24,(8,4,4,2):21,(8,4,2,2,2):36,(8,2,2,2,2,2):56,
      (6,6,6):10,(6,6,4,2):20,(6,6,2,2,2):36,(6,4,4,4):20,
      (6,4,4,2,2):30,(6,4,2,2,2,2):48,(6,2,2,2,2,2,2):68,
      (4,4,4,4,2):30,(4,4,4,2,2,2,2):40,(4,4,2,2,2,2,2,2):60,
      (4,2,2,2,2,2,2,2,2):80,(2,2,2,2,2,2,2,2,2,2):100}

d, L = 9, evens(18)
assert len(L) == 30 and sorted(L) == sorted(Y)
one = [(a, b) for a in L for b in L if a != b and sd(a, b)]
print(len(one), min(Y[a] - Y[b] for a, b in one))    # 417 0

blk = {}
for r in range(d + 1):
    for s in range(d//2 + 1):
        V = [(al, la) for al in alphas(r, s, d)
              for la in evens(d - sum(al))] if r + s else []
        if V:
            blk[(r, s)] = V
print(len(blk), sum(len(V) for V in blk.values()))    # 15 131

A = lambda u: key([2*a for a in u[0]] + 2*list(u[1]))

```

```

def Bs(u):
    t = sorted([2*a for a in u[0]], reverse=True)
    return key([t[0] + t[1]] + t[2:] + 2*list(u[1]))
C = lambda u, v: key([a + b for a, b in zip(u[0], v[0])]
                    + list(u[1]) + list(v[1]))

dg = [(A(u), Bs(u)) for (r, s), V in blk.items()
      if r + s >= 2 for u in V]
of = [(A(u), A(v), C(u, v)) for V in blk.values()
      for u, v in combinations(V, 2)]
print(len(dg), len(of), len(dg) + len(of))          # 119 937 1056
print(min(Y[a] - Y[b] for a, b in dg),
      min(Y[a] + Y[b] - 2*Y[c] for a, b, c in of))  # 0 0
print(all(sd(a, b) for a, b in dg),
      all(sd(a + b, c + c) for a, b, c in of))      # True True

two = [(a, b, m) for a, b in cwr(L, 2) for m in L
      if sd(a + b, m + m)]
bad = [t for t in two if Y[t[0]] + Y[t[1]] - 2*Y[t[2]] < 0]
print(len(two), bad)
# 7025 [((10, 8), (4, 4, 4, 2, 2, 2), (8, 4, 4, 2))]
print(Y[(10,8)] + Y[(4,4,4,2,2,2)] - 2*Y[(8,4,4,2)]) # -2

```

REFERENCES

- [1] J. Acevedo, G. Blekherman, S. Debus, and C. Riener, *Symmetric nonnegative functions, the tropical Vandermonde cell and superdominance of power sums*, Collect. Math. **77** (2026), 555–600, doi:10.1007/s13348-025-00481-z; arXiv:2408.04616.